

Report
Of
Artificial Intelligence Internship

Submitted
To
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
MCA



By
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CANDIDATE'S DECLARATION

I, VAIBHAV PATHAK, do hereby declare that the internship report titled "Artificial Intelligence Internship" has been completed by me to fulfil the requirement for the award of the degree of Master of Computer Applications (MCA). All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own work and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature

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ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Artificial Intelligence Internship Program at Codec Technologies. I am thankful for their aspiring guidance, invaluable constructive criticism, and friendly advice during the course of this internship. I am sincerely grateful to the mentors and trainers at Codec Technologies for sharing their truthful and illuminating views on a number of issues related to Artificial Intelligence and Machine Learning, and for structuring the program in a way that allowed hands-on, practical learning.

I would also like to express my sincere thanks to the faculty of the School of Computer Science and Engineering, IILM University, Greater Noida, for their continuous support and for providing me the opportunity to undertake this internship as part of my MCA curriculum.

Finally, I express my heartfelt gratitude to my parents and friends for their constant encouragement and blessings throughout this journey.

Signature

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INTERNSHIP COMPLETION CERTIFICATE



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4. Project Description

4.1 Introduction

Artificial Intelligence (AI) has emerged as one of the most transformative technologies of the current decade, reshaping the way industries approach problem-solving, automation, and decision-making. As a Master of Computer Applications (MCA) student with a background in full-stack development (JavaScript, Node.js, React, and MongoDB), I undertook a one-month Artificial Intelligence Internship Program with Codec Technologies from 27 July 2026 to 27 August 2026 to build a strong practical foundation in AI, Machine Learning (ML), and Deep Learning (DL), and to apply these concepts to real, working projects.

The internship was structured as a progressive, four-week curriculum that moved from the fundamentals of Python-based data handling to classical machine learning, then into neural networks and deep learning, and finally into advanced applications such as Natural Language Processing (NLP) and Reinforcement Learning (RL), culminating in an independent capstone project. This report documents the concepts covered, the skills developed, the practical work undertaken, and the outcomes achieved during the internship.

The internship gave me the opportunity to move beyond theoretical knowledge of AI and actually build, train, evaluate, and refine machine learning models end-to-end - from raw data to a working, testable application - which has meaningfully strengthened my understanding of how AI systems are designed and deployed in practice.

4.2 Organization Profile

Codec Technologies (<https://codectechnologies.in/>) is a technology training and skill-development organization that offers structured, industry-oriented internship and certification programs for students and early-career professionals in emerging technology domains, including Artificial Intelligence, Machine Learning, Data Science, and related fields. The organization focuses on bridging the gap between academic coursework and industry-relevant, hands-on technical skills through mentor-guided, project-based learning.

As part of its Artificial Intelligence Internship Program, Codec Technologies provides a structured curriculum, learning resources, and mentorship to help interns build practical competence in Python-based AI development, covering data handling, classical machine learning, deep learning, and advanced applications such as NLP, supported by a capstone project component to consolidate learning.

4.3 Problem Statement

Despite growing exposure to AI concepts through coursework and self-study, translating theoretical understanding of Machine Learning and Deep Learning into working, deployable solutions remains a significant challenge for most students. Common gaps include difficulty in preparing and cleaning real-world data, choosing the right model for a given problem, correctly evaluating model performance, and structuring a complete AI project from data ingestion through to a usable output.

The internship was designed to address this gap directly: rather than studying AI concepts in isolation, the program required implementing each concept as working code, and concluded with an independent capstone project that involved identifying a real classification problem, sourcing and preparing data, building and training a model, evaluating its performance, and packaging it as a usable tool.

4.4 Project Objectives

- To understand the foundational concepts of Artificial Intelligence, Machine Learning, and Deep Learning, and how they differ in scope and application.
- To gain hands-on proficiency in Python libraries used for AI development, including NumPy, Pandas, Matplotlib, Seaborn, scikit-learn, TensorFlow, and Keras.
- To learn and apply core data preprocessing techniques, including data cleaning, normalization, and handling of missing values.
- To implement and evaluate supervised learning models (linear regression, logistic regression) using standard performance metrics.
- To build and train simple neural networks and Convolutional Neural Networks (CNNs) for tasks such as image classification.

- To explore Natural Language Processing (NLP) techniques for text classification and sentiment analysis, and to understand the basics of Reinforcement Learning (RL).
- To independently design, build, train, and evaluate a complete capstone AI project, and to document and present the work professionally.

4.5 Scope of the Project

The scope of the internship covered the complete beginner-to-intermediate AI development pipeline over four structured weeks: (1) Python foundations and data handling; (2) classical machine learning with supervised learning algorithms; (3) neural networks and deep learning, including CNNs for image data; and (4) advanced applications in NLP and Reinforcement Learning, concluding with a capstone project. Within this scope, the practical, hands-on component of the internship focused specifically on two applied machine learning projects built end-to-end: a Handwritten Digit Recognizer (an image classification task using neural networks, connecting to the Week 3 deep learning/CNN module) and a Spam Message Classifier (a text classification task using Natural Language Processing techniques, connecting to the Week 4 NLP module). These two projects together cover both major branches of applied AI explored during the internship - vision-based classification and text-based classification - and served as the practical demonstration of concepts learned across all four weeks.

The scope did not extend to production-grade deployment (e.g., hosting the models as live web services with continuous training pipelines), as the internship's focus was on building strong conceptual and practical foundations within the one-month duration.

4.6 Technologies and Tools Used

Table 4.1: Technologies and tools used during the internship

Category	Tools / Technologies
Programming Language	Python 3
Data Handling	NumPy, Pandas
Data Visualization	Matplotlib, Seaborn
Classical Machine Learning	scikit-learn (Logistic/Linear Regression, Naive Bayes, MLPClassifier)
Deep Learning	TensorFlow, Keras (for neural networks and CNNs)
Natural Language	NLTK, SpaCy, TF-IDF Vectorization

Processing	
Model Persistence	Joblib
Development Environment	Python virtual environments (venv), command-line tools
Version Control	Git and GitHub

4.7 System Architecture

Both practical projects developed during the internship follow a standard supervised machine learning pipeline, which can be summarized as: Data Input -> Data Preprocessing -> Feature Extraction -> Model Training -> Model Evaluation -> Prediction/Output. Figure 4.1 illustrates this general architecture as applied to the two projects.

For the Handwritten Digit Recognizer: raw pixel data of 8x8 grayscale digit images is first scaled using a StandardScaler, then fed into a Multi-Layer Perceptron (MLP) neural network with two hidden layers (64 and 32 neurons, ReLU activation), which outputs a predicted digit class (0-9).

For the Spam Message Classifier: raw text messages are converted into numerical TF-IDF (Term Frequency-Inverse Document Frequency) feature vectors after stop-word removal, which are then passed to a Multinomial Naive Bayes classifier that outputs a predicted label ("spam" or "ham").

Figure 4.1: General architecture - Raw Data -> Preprocessing -> Feature Extraction -> Model Training -> Evaluation -> Prediction Output

4.8 Methodology

The internship followed a structured, four-week methodology combining conceptual learning with immediate hands-on implementation, as summarised below.

4.8.1 Week 1: Foundations of AI and Python for Data Handling

- Studied the core distinctions between Artificial Intelligence, Machine Learning, and Deep Learning, along with real-world industry applications and types of AI.
- Practised Python fundamentals - data structures, control flow, and functions - with a focus on writing clean, reusable code.

- Learned data manipulation using NumPy and Pandas, including data cleaning, normalization, and handling of missing values.
- Practised basic data visualization using Matplotlib and Seaborn to understand data distributions before modelling.

4.8.2 Week 2: Machine Learning Basics

- Implemented linear regression for continuous-value prediction and logistic regression for binary classification using scikit-learn.
- Learned to evaluate model performance using accuracy, precision, recall, and F1-score, and to interpret confusion matrices.
- Applied train-test splitting, cross-validation, and basic hyperparameter tuning to improve model generalisation.

4.8.3 Week 3: Introduction to Neural Networks and Deep Learning

- Studied the fundamentals of neural networks - layers, weights, activation functions, and backpropagation.
- Built and trained a simple neural network using TensorFlow/Keras concepts, and separately implemented an MLPClassifier (scikit-learn) for digit recognition.
- Learned the role of Convolutional Neural Networks (CNNs) in image recognition and how convolution/pooling layers extract spatial features.
- Applied this learning directly by building the Handwritten Digit Recognizer project (detailed below), training a neural network to classify digits 0-9 with approximately 97% test accuracy.

4.8.4 Week 4: Advanced AI Applications and Capstone Project

- Studied NLP fundamentals - text preprocessing, tokenization, and sentiment analysis - along with an introduction to Reinforcement Learning and Q-learning principles.
- Selected the Spam Message Classifier as one of the two capstone-style projects: prepared a labelled spam/ham text dataset, converted text to TF-IDF feature

vectors, and trained a Multinomial Naive Bayes classifier, achieving reliable classification performance on unseen messages.

- Documented both projects with README files, requirements files, and reusable, well-commented scripts, and packaged them for submission via [GitHub](#), in line with standard project-delivery practice.

4.8.5 Practical Project Summaries

Project 1 - Handwritten Digit Recognizer: A neural network (MLPClassifier with two hidden layers) trained on an 8x8-pixel digit image dataset. The pipeline includes feature scaling, model training, evaluation via classification report and confusion matrix, and visualization of sample predictions. The trained model achieved approximately 97% accuracy on the held-out test set.

GitHub Repository: <https://github.com/Vaibhav007-code/INTERNSHIP-PROJECT>

Project 2 - Spam Message Classifier: A Natural Language Processing pipeline that converts raw text messages into TF-IDF feature vectors and classifies them as "spam" or "ham" using a Multinomial Naive Bayes model. The system includes a command-line interface for classifying new, user-supplied messages with a confidence score, in addition to full training and evaluation on a labelled dataset.

GitHub Repository: <https://github.com/Vaibhav007-code/INTERNSHIP-PROJECT> **4.9**

Expected Outcomes

- A solid conceptual understanding of Artificial Intelligence, Machine Learning, and Deep Learning, and how they relate to one another.
- Practical proficiency in the Python AI/ML ecosystem, including NumPy, Pandas, scikit-learn, TensorFlow, and Keras.
- The ability to independently take a machine learning problem from raw data through preprocessing, model training, evaluation, and deployment as a usable script/tool.
- Two complete, working, well-documented AI projects (an image classifier and a text classifier) that can be showcased in a technical portfolio or GitHub profile.
- Improved understanding of model evaluation practices (accuracy, precision, recall, F1-score, confusion matrices) and how to interpret them to improve a model.

- Stronger foundation for pursuing more advanced AI/ML coursework, projects, or roles in the future, complementing existing full-stack development skills.

4.10 Certificates of Completion and Other Communication Mail Proof



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